Experiment 4 : Data Wrangling on Real Estate Marke

Aim: Data Wrangling on Real Estate Market. The dataset contains information about housing pri real estate market, with attributes such as property characteristics, location and sale prices. The to perform data wrangling to gain insights into the factors influencing housing prices and prepare dataset for further analysis or modeling.

Objective: To perform data wrangling operations using Python on the real estate dataset.

In [1]:
```
# A52 Dipanshu Ambilkar
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [2]:
```
# Step 1: Import the RealEstate dataset
df = pd.read_csv("realestate.csv")
print("Shape:", df.shape)
print("Original columns:", list(df.columns))
df.head()
```

```
Shape: (250, 7)
Original columns: ['Property ID', 'Neighborhood ', 'Property-Type', 'Area (sq
' Bedrooms', 'Sale Price (Rs)', 'Year Built']
```

Out[2]:

	Property ID	Neighborhood	Property-Type	Area (sqft)	Bedrooms	Sale Price (Rs)	Ye
0	1	Wakad	Villa	604	1	10898035	2
1	2	Aundh	Villa	2153	3	16490355	2
2	3	Kothrud	Villa	2912	3	16767181	2
3	4	Hinjewadi	Flat	1239	2	85909200	1
4	5	Aundh	Flat	2065	3	17276029	2

In [3]:
```
# Clean column names: remove spaces and special characters
df.columns = df.columns.str.strip().str.lower().str.replace(" ", "_").str
print("Cleaned columns:", list(df.columns))
df.head()
```

```
Cleaned columns: ['property_id', 'neighborhood', 'property_type', 'area_sqft'
'bedrooms', 'sale_price_rs', 'year_built']
```

Out[3]:

	property_id	neighborhood	property_type	area_sqft	bedrooms	sale_price_rs
0	1	Wakad	Villa	604	1	10898035
1	2	Aundh	Villa	2153	3	16490355
2	3	Kothrud	Villa	2912	3	16767181
3	4	Hinjewadi	Flat	1239	2	85909200
4	5	Aundh	Flat	2065	3	17276029

In [4]:

```
# Step 2: Check missing values in the dataset
print(df.isnull().sum())
print("Total missing values:", df.isnull().sum().sum())
```

```
property_id      0
neighborhood     0
property_type    0
area_sqft        0
bedrooms         0
sale_price_rs    0
year_built       0
dtype: int64
Total missing values: 0
```

In [5]:
```
# Handle missing values: fill numeric columns with their median
for col in ["area_sqft", "sale_price_rs"]:
    df[col] = df[col].fillna(df[col].median())
print("Missing values after handling:", df.isnull().sum().sum())
```

```
Missing values after handling: 0
```

In [6]:
```
# Step 3: Merge with additional neighborhood information
neighborhood_info = pd.DataFrame({
    "neighborhood": ["Wakad", "Aundh", "Kothrud", "Hinjewadi"],
    "zone": ["West", "West", "Central", "West"],
    "area_type": ["Residential", "Residential", "Residential", "IT Corrid
})
df = pd.merge(df, neighborhood_info, on="neighborhood", how="left")
print("Shape after merging:", df.shape)
df.head()
```

```
Shape after merging: (250, 9)
```

Out[6]:

	property_id	neighborhood	property_type	area_sqft	bedrooms	sale_price_rs
0	1	Wakad	Villa	604	1	10898035
1	2	Aundh	Villa	2153	3	16490355
2	3	Kothrud	Villa	2912	3	16767181
3	4	Hinjewadi	Flat	1239	2	8590920
4	5	Aundh	Flat	2065	3	17276029

In [7]:
```
# Step 4: Filter and subset the data - only Flats
flats = df[df["property_type"] == "Flat"]
print("Total properties:", len(df))
print("Flats only:", flats.shape)
flats.head()
```

```
Total properties: 250
Flats only: (82, 9)
```

Out[7]:

	property_id	neighborhood	property_type	area_sqft	bedrooms	sale_price_rs
3	4	Hinjewadi	Flat	1239	2	8590920
4	5	Aundh	Flat	2065	3	17276029
5	6	Aundh	Flat	2892	4	19570342
6	7	Aundh	Flat	2014	2	13812021
8	9	Kothrud	Flat	983	3	7719614

In [8]:
```
# Step 5: Identify and handle outliers in sale price using IQR method
q1 = df["sale_price_rs"].quantile(0.25)
q3 = df["sale_price_rs"].quantile(0.75)
iqr = q3 - q1
upper = q3 + 1.5 * iqr
print("Upper limit for sale price:", round(upper, 2))
print("Outlier rows:", int((df["sale_price_rs"] > upper).sum()))
df = df[df["sale_price_rs"] <= upper]
print("Shape after removing outliers:", df.shape)
```

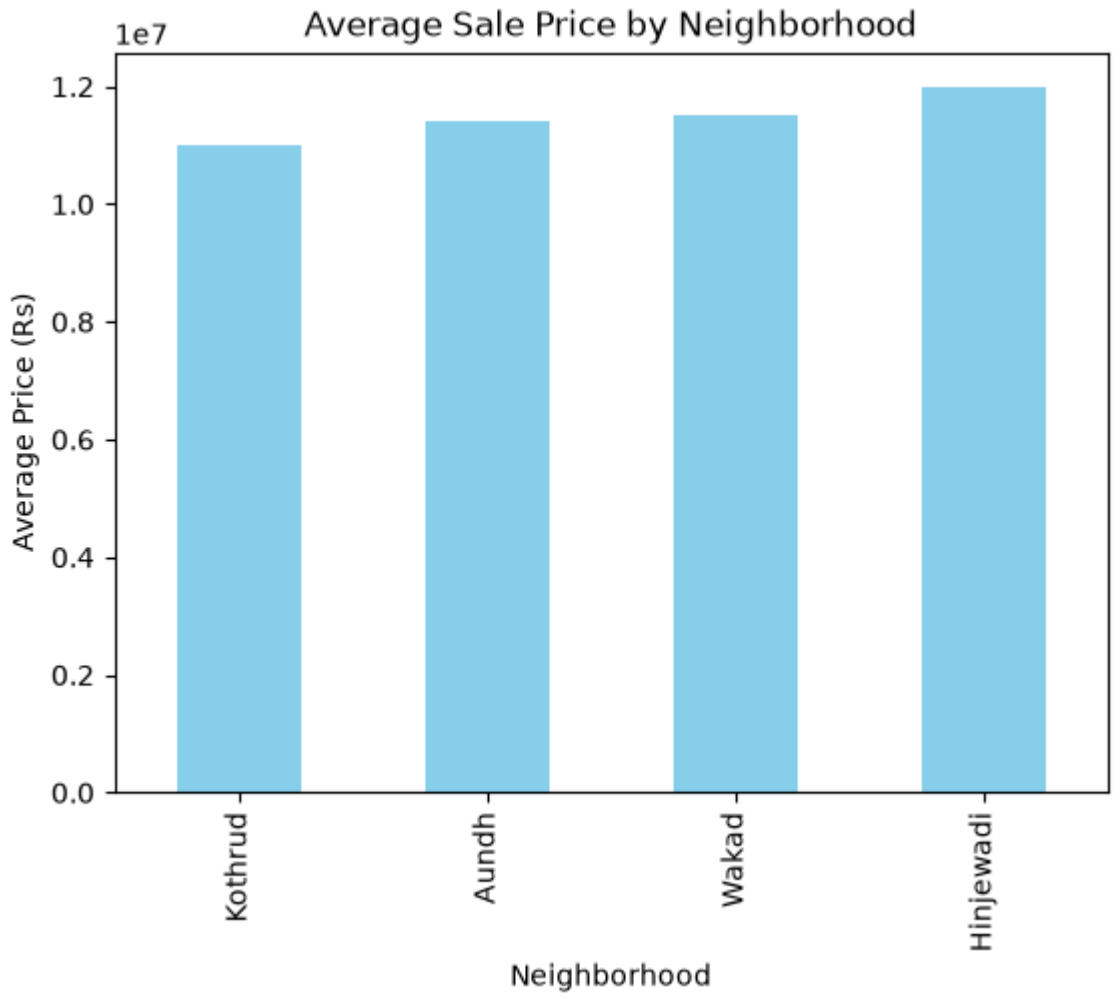
```
Upper limit for sale price: 28514656.25
Outlier rows: 0
Shape after removing outliers: (250, 9)
```

In [9]:
```
# Step 6: Aggregate data - average sale price by neighborhood and propert
avg_by_neighborhood = df.groupby("neighborhood")["sale_price_rs"].mean().
print("Average sale price by neighborhood:")
print(avg_by_neighborhood)
print()
print("Average sale price by property type:")
print(df.groupby("property_type")["sale_price_rs"].mean())
```

```
Average sale price by neighborhood:
neighborhood
Kothrud      1.100382e+07
Aundh        1.142568e+07
Wakad        1.151917e+07
Hinjewadi    1.197643e+07
Name: sale_price_rs, dtype: float64

Average sale price by property type:
property_type
Flat     1.214124e+07
Plot     1.136373e+07
Villa    1.091508e+07
Name: sale_price_rs, dtype: float64
```

In [10]:
```
# Bar plot of average sale price by neighborhood
avg_by_neighborhood.plot(kind="bar", color="skyblue")
plt.title("Average Sale Price by Neighborhood")
plt.xlabel("Neighborhood")
plt.ylabel("Average Price (Rs)")
plt.show()
```



In [11]:
```
# Step 7: Encode categorical variables using one-hot encoding
df_encoded = pd.get_dummies(df, columns=["neighborhood", "property_type"])
print("Shape after encoding:", df_encoded.shape)
df_encoded.head()
```

```
Shape after encoding: (250, 12)
```

Out[11]:

	property_id	area_sqft	bedrooms	sale_price_rs	year_built	zone	area_ty
0	1	604	1	10898035	2020	West	Resident
1	2	2153	3	16490355	2003	West	Resident
2	3	2912	3	16767181	2021	Central	Resident
3	4	1239	2	8590920	1991	West	IT Corric
4	5	2065	3	17276029	2008	West	Resident

Conclusion: In this way we performed data wrangling on the real estate dataset: cleaned colum names, handled missing values, merged additional neighborhood data, filtered subsets, removed outliers, aggregated average prices and encoded categorical variables, preparing the dataset for analysis or modeling.